

H bridge

An **H-bridge** is an electronic circuit that allows you to control the direction of a DC motor by using four transistors arranged in an “H” shape. Think of it as a smart switchboard that can reverse the flow of current through the motor, making it spin forward or backward.

Here’s how it works in transistor terms:

- The four transistors are placed in two pairs: two on the high side (connected to the positive voltage) and two on the low side (connected to ground).
- To make the motor spin in one direction, you turn on one high-side transistor and the opposite low-side transistor diagonally. This lets current flow from the power supply, through the motor, and to ground in a specific path.
- To reverse the motor, you switch on the other high-side transistor and the opposite low-side transistor. This reverses the current flow through the motor, making it spin the other way.
- Importantly, you **never turn on both transistors on the same side at the same time**, because that would create a short circuit between power and ground, damaging the circuit.
- By controlling which transistors are on or off with input signals (like from a microcontroller), you can precisely control motor direction and even stop or brake it by turning off certain transistor pairs or turning on both low or high sides.

the H-bridge uses transistor switches to carefully control current paths, allowing the motor to run forward, backward, or stop, all with just four transistor switches. This makes it essential for robotics, motor drivers, and many electronics projects that need reversible motor control.